

SRI VENKATESWARA UNIVERSITY, TIRUPATI

Revised Curriculum for MCA Programme with Effect from 2025–2026

MCA 301: COMPUTER NETWORKS

COMPLETE UNIVERSITY EXAMINATION STUDY MATERIAL

Units I–V

Scope: Data Communication, Physical Layer, Data Link Layer, Network Layer, Transport Layer, Security and Application Layer.

Academic Edition – Revised Curriculum 2025–2026

Structured for classroom support, self-study, numerical practice and university examination preparation

September 2026

Academic Purpose and Quality Statement

This material is a comprehensive student study resource aligned to the MCA 301: Computer Networks syllabus supplied for the Sri Venkateswara University Revised Curriculum 2025–2026. It is designed to support teaching, revision and examination preparation. It does not replace the university syllabus, prescribed regulations or faculty instruction.

Important Exam Point

Technical accuracy policy: Core Internet protocol details are aligned with IETF RFCs where applicable, while foundational networking explanations follow standard university-level Computer Networks terminology and the recommended textbooks listed in the syllabus. Where older textbook conventions can differ from current protocol specifications, the material identifies the distinction.

How to Use This Material

For every major concept:

1. Read the definition and purpose.
2. Study the diagram/format.
3. Follow the step-by-step operation.
4. Work the numerical example if present.
5. Memorize the exam points and comparison table.
6. Practice the corresponding 2-, 5- and 10-mark questions.

Syllabus Coverage Matrix

Definition

This matrix maps the prescribed MCA 301 syllabus for the Revised Curriculum 2025–2026 to the corresponding study-material sections. Unit IV follows the detailed 18-lesson structure; Units I, II, III and V are organized into focused lessons covering every syllabus item.

Unit	Prescribed syllabus area	Material coverage
I	Data Communication; networks; topologies; OSI/TCP-IP; physical layer; signals; impairment; performance; FDM/WDM/TDM; media; switching	Lessons I-1 to I-7
II	Data link design; framing; flow/error control; parity/checksum/CRC/Hamming; 802.11; Bluetooth; Stop-and-Wait/GBN/SR ARQ; Frame Relay; ATM	Lessons II-1 to II-5
III	IPv4/IPv6; transition; ARP/RARP/DHCP; ICMP/IGMP; unicast routing; RIP/OSPF; link state/distance vector; DVMRP/MO-SPF	Lessons III-1 to III-7
IV	Process-to-process; UDP; TCP; flow/error/-congestion; QoS; security; cryptography; message security; signatures; authentication; e-mail; web; social issues	Lessons IV-1 to IV-18
V	DNS; e-mail architecture/services; POP3; SMTP; FTP; WWW; HTTP	Lessons V-1 to V-5

Important Exam Point

Parity appears in Unit II as prescribed. A separate parity revision can also be used while studying Unit IV error-control concepts, but parity is not a TCP/UDP checksum mechanism.

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Part I

UNIT I – DATA COMMUNICATION AND PHYSICAL LAYER

Chapter 1

Data Communication

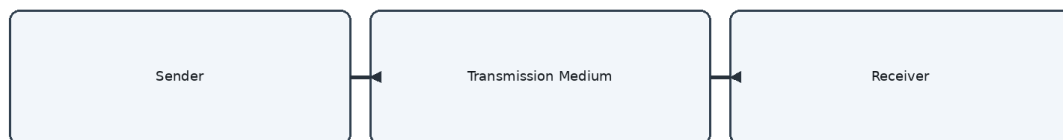
Definition

Learning objectives: Define data communication, identify its components, explain data representation and distinguish data-flow modes.

1.1 Definition and Purpose

Data communication is the exchange of data between two or more devices through a transmission medium using agreed communication rules (protocols). A basic communication system contains a sender, receiver, message, transmission medium and protocol.

Basic Data Communication Model



1.2 Components

- **Message:** information to be communicated.
- **Sender:** device that originates the message.
- **Receiver:** device that accepts the message.
- **Medium:** physical or wireless path.
- **Protocol:** rules governing communication.

1.3 Data Representation

Text is represented using character encodings; numbers may be represented in binary; images are represented as pixels/samples; audio and video are represented as sampled/encoded data. In networking, all are ultimately carried as bits.

1.4 Data Flow

Simplex permits one-way communication; **half-duplex** permits two-way communication but not simultaneously; **full-duplex** permits simultaneous two-way communication.

Important Exam Point

Exam sequence: components → data representation → data-flow modes → example.

2-Mark Answer

Data communication is the exchange of data between communicating devices through a medium according to a protocol.

5-Mark Answer

Define data communication and explain the five components and three data-flow modes with examples.

10-Mark Answer

Explain data communication, components, representation and flow modes with a labeled diagram and real-world examples.

Chapter 2

Networks: Distributed Processing, Criteria, Physical Structure and Categories

Definition

Learning objectives: Explain distributed processing, network criteria, physical structures and common network categories.

2.1 Distributed Processing

In distributed processing, computational tasks and resources are distributed among multiple interconnected systems rather than being performed by one central computer. Communication networks enable resource sharing and coordination.

2.2 Network Criteria

Important criteria include **performance**, **reliability** and **security**. Performance can be considered using throughput and delay; reliability includes failure frequency and recovery; security concerns protection against unauthorized access and misuse.

2.3 Physical Structure

A network may be described by point-to-point links or multipoint/shared links. Physical structure also considers how devices are interconnected.

2.4 Categories by Geographic Scope

- **PAN:** personal area.
- **LAN:** local area.
- **MAN:** metropolitan area.
- **WAN:** wide area.

Important Exam Point

Do not define LAN, MAN and WAN only by an exact distance. Their meanings are primarily based on geographic scope and typical deployment.

2-Mark Answer

A LAN connects devices over a relatively limited geographic area such as a building or campus.

5-Mark Answer

Explain network criteria and network categories with examples.

10-Mark Answer

Discuss distributed processing, performance/reliability/security criteria, physical structures and LAN/MAN/WAN categories.

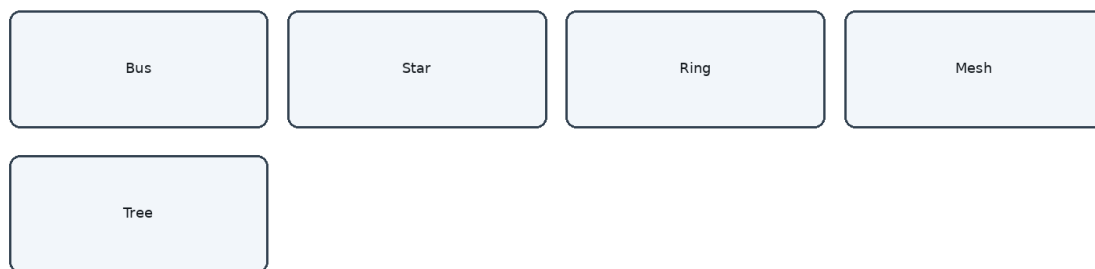
Chapter 3

Network Topologies

Definition

Learning objectives: Explain bus, star, ring, mesh and tree topologies with advantages and limitations.

Common Network Topologies



3.1 Bus

All devices share a common backbone. It can be economical but a backbone fault can affect communication and shared-medium contention must be managed.

3.2 Star

Each node connects to a central device. It is easy to manage and isolate a single edge-link fault, but failure of the central device can affect many nodes.

3.3 Ring

Each node has a logical/physical ring relationship. Ring implementations differ in access control and fault behavior.

3.4 Mesh

Nodes have many interconnections. Full mesh offers multiple paths but requires many links; partial mesh reduces cost while retaining redundancy.

3.5 Tree

Hierarchical interconnection of star-like segments. It supports structured expansion but higher-level failures can affect branches.

2-Mark Answer

Topology is the arrangement of nodes and links in a network.

5-Mark Answer

Explain any three network topologies with diagrams, advantages and limitations.

10-Mark Answer

Compare bus, star, ring, mesh and tree topologies with diagrams and suitable applications.

Chapter 4

OSI Reference Model and TCP/IP Model

Definition

Learning objectives: Explain the OSI layers, responsibilities and relationship to the TCP/IP model.

OSI Reference Model



4.1 OSI Layers

1. Application – network services to applications.
2. Presentation – representation, translation, compression and related functions.
3. Session – session establishment/management concepts.
4. Transport – process-to-process delivery and transport services.
5. Network – logical addressing and routing.
6. Data Link – framing, link control and error detection/correction functions.
7. Physical – transmission of bits/signals.

4.2 TCP/IP Model

The Internet architecture is commonly described using application, transport, Internet/network, and link/network-access functions. Textbooks may present four or five layers depending on how the link/physical functions are divided.

4.3 Encapsulation

As data moves downward, each layer adds its control information; at the receiver the process is reversed through decapsulation.

Important Exam Point

Do not claim that every real protocol maps one-to-one to one OSI layer. Models are conceptual frameworks; actual protocols may span functions.

2-Mark Answer

The OSI model has seven layers from Physical to Application.

5-Mark Answer

List the seven OSI layers and one or two major functions of each.

10-Mark Answer

Explain OSI and TCP/IP models, layer functions, encapsulation and the major differences in architectural presentation.

Chapter 5

Physical Layer: Data and Signals

Definition

Learning objectives: Distinguish analog/digital data and signals and explain basic signal parameters.

5.1 Data and Signals

Data is information; a **signal** is the physical/electromagnetic representation used to carry that information.

- Analog data varies continuously; digital data is represented using discrete values.
- Analog signals vary continuously; digital signals use discrete signal levels/waveforms.

5.2 Basic Signal Concepts

Amplitude describes magnitude; frequency is cycles per second (Hz); period is $T = 1/f$; phase describes relative timing/position of a periodic waveform.

5.3 Bit Rate and Baud Rate

Bit rate is bits per second. Baud rate is signal elements per second. They are equal only under particular signaling conditions; one signal element can encode multiple bits.

Numerical Example

If a periodic signal has frequency 2 MHz:

$$T = \frac{1}{2 \times 10^6} = 0.5 \mu s.$$

2-Mark Answer

Frequency is the number of cycles of a periodic signal per second.

5-Mark Answer

Differentiate analog/digital data and signals and define amplitude, frequency, period and phase.

10-Mark Answer

Explain data, signals, signal parameters and bit/ baud rate with numerical examples.

Chapter 6

Transmission Impairment and Performance

Definition

Learning objectives: Explain attenuation, distortion, noise, bandwidth, throughput, propagation delay and transmission delay.

6.1 Transmission Impairments

- **Attenuation:** loss of signal strength.
- **Distortion:** change in signal shape because frequency components are affected differently.
- **Noise:** unwanted energy that alters the received signal.

6.2 Performance Measures

Bandwidth is a frequency range or, in a digital context, may refer to capacity depending on context. **Throughput** is the actual achieved data rate. **Propagation delay** is time for the signal to travel through the medium:

$$d_{prop} = \frac{\text{distance}}{\text{propagation speed}}.$$

Transmission delay is time to place all bits of a packet on the link:

$$d_{trans} = \frac{\text{packet size in bits}}{\text{data rate}}.$$

Numerical Problem

A 1,000,000-bit packet is transmitted at 10 Mbps:

$$d_{trans} = \frac{10^6}{10 \times 10^6} = 0.1s.$$

For 20 km at 2×10^8 m/s:

$$d_{prop} = \frac{20,000}{2 \times 10^8} = 100 \mu s.$$

Common Mistake

Propagation delay and transmission delay are different. Transmission delay depends on packet size and rate; propagation delay depends on distance and propagation speed.

2-Mark Answer

Attenuation is reduction in signal strength as it travels.

5-Mark Answer

Explain the three major impairments and distinguish propagation and transmission delay.

10-Mark Answer

Explain impairments and performance measures with formulas and numerical examples.

Chapter 7

Multiplexing, Transmission Media and Switching

Definition

Learning objectives: Explain FDM, WDM, TDM, guided/unguided media and circuit/packet switching.

Multiplexing Concept



7.1 Multiplexing

Multiplexing allows multiple signals/flows to share a common medium.

- **FDM:** users occupy different frequency bands simultaneously.
- **WDM:** optical wavelengths carry separate channels over fiber.
- **TDM:** users share the channel at different time slots.

7.2 Transmission Media

Guided: twisted-pair cable, coaxial cable and fiber-optic cable. **Unguided:** radio waves, microwaves and infrared.

7.3 Switching

Switching Techniques



Circuit switching establishes a dedicated path/resources for a communication session. **Packet switching** divides data into packets that share network resources; it may use datagram or virtual-circuit approaches.

2-Mark Answer

FDM separates channels by frequency; TDM separates them by time; WDM separates optical channels by wavelength.

5-Mark Answer

Explain FDM, WDM and TDM with applications.

10-Mark Answer

Discuss multiplexing, guided/unguided media and circuit versus packet switching.

Part II

UNIT II – DATA LINK LAYER

Chapter 8

Data Link Layer: Framing and Flow Control

Definition

Learning objectives: Explain framing, flow control and the basic responsibilities of the data link layer.

Framing Concept



8.1 Data Link Layer

The data link layer provides node-to-node delivery over a link and commonly handles framing, link addressing, error detection and flow/access control depending on the technology.

8.2 Framing

Framing divides a bit stream into identifiable frames. Common approaches include character/byte-oriented framing, byte stuffing, bit stuffing and physical-layer coding violations in suitable systems.

8.3 Flow Control

Flow control coordinates sender and receiver so the sender does not overwhelm the receiver. Stop-and-wait sends one frame and waits; sliding-window protocols allow multiple outstanding frames.

2-Mark Answer

Framing is the process of dividing a bit stream into manageable data-link frames.

5-Mark Answer

Explain framing methods and the purpose of flow control.

10-Mark Answer

Explain data-link design issues with emphasis on framing and flow control, including examples.

Chapter 9

Error Detection and Correction: Parity, Checksum, CRC and Hamming Code

Definition

Learning objectives: Solve parity/checksum/CRC/Hamming examples and distinguish detection from correction.

Error Detection Methods



9.1 Parity

A parity bit is added so the number of 1s is even or odd. It is simple but has limited detection capability.

Example

Data = 1011001 has four 1s. Even parity bit = 0; odd parity bit = 1.

9.2 Checksum

Words are added using one's-complement arithmetic and the final complement is transmitted. Receiver verification is performed using the corresponding one's-complement rule.

9.3 CRC

CRC treats the bit string as a polynomial over GF(2). The sender appends zeros, divides by the generator polynomial using modulo-2 division and appends the remainder. The receiver divides the received codeword; a nonzero remainder indicates an error under the CRC model.

9.4 Hamming Code

Hamming codes use parity positions and can provide error detection/correction capability according to the specific code construction. For a basic single-error-correcting Hamming code, the number r of parity bits must satisfy:

$$2^r \geq m + r + 1$$

where m is the number of data bits.

2-Mark Answer

CRC uses polynomial/modulo-2 division to generate a remainder used for error detection.

5-Mark Answer

Compare parity, checksum and CRC.

10-Mark Answer

Explain parity, checksum, CRC and Hamming code, including at least one worked numerical example.

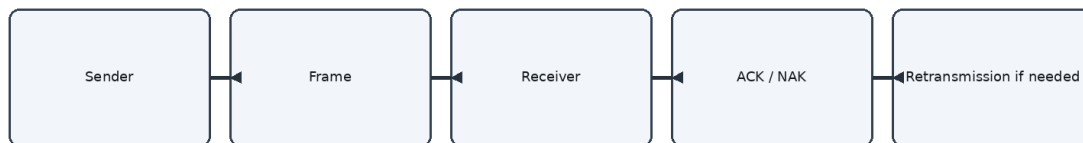
Chapter 10

Data Link Protocols: Stop-and-Wait, Go-Back-N and Selective Repeat ARQ

Definition

Learning objectives: Explain ARQ operation, ACK/timeout behavior and compare Stop-and-Wait, Go-Back-N and Selective Repeat.

ARQ Reliability Concept



10.1 ARQ

Automatic Repeat reQuest uses acknowledgments, sequence numbers and retransmission to recover from lost/corrupted frames.

10.2 Stop-and-Wait ARQ

The sender transmits one frame and waits for an ACK. On timeout or negative indication it retransmits. Sequence numbering is used to distinguish duplicates.

10.3 Go-Back-N

The sender may transmit multiple frames. The receiver generally uses cumulative acknowledgments and may discard out-of-order frames; after a loss, the sender retransmits the missing frame and

subsequent outstanding frames.

10.4 Selective Repeat

The receiver can buffer out-of-order frames and acknowledge them selectively. The sender retransmits only missing/unacknowledged frames.

	Go-Back-N	Selective Repeat
Receiver buffering	Limited in classic model	Buffers out-of-order frames
Retransmission	Lost frame and following outstanding frames	Only missing frames
Efficiency	Lower when errors are frequent	Higher but more complex

2-Mark Answer

ARQ means Automatic Repeat reQuest and uses feedback plus retransmission for reliability.

5-Mark Answer

Explain Stop-and-Wait ARQ and Go-Back-N.

10-Mark Answer

Compare Stop-and-Wait, Go-Back-N and Selective Repeat ARQ with timing examples.

Chapter 11

Wireless LANs: IEEE 802.11 and Bluetooth

Definition

Learning objectives: Describe the syllabus-level concepts of IEEE 802.11 wireless LANs and Bluetooth.

Wireless LAN Technologies



11.1 IEEE 802.11

IEEE 802.11 specifies wireless LAN technologies. At the syllabus level, important concepts include wireless stations, access points, association, shared-medium access and frame-based communication. CSMA/CA is used rather than Ethernet-style collision detection because a wireless station generally cannot reliably detect a collision while transmitting.

11.2 Bluetooth

Bluetooth is a short-range wireless technology family used for personal-area communication and device connectivity. It is designed for low-power, short-range links and supports device discovery, pairing and different profiles.

Common Mistake

Do not equate Wi-Fi and Bluetooth. They serve overlapping but different deployment and use cases and are based on different standards/families.

2-Mark Answer

IEEE 802.11 is the major IEEE wireless LAN standard family; Bluetooth is a short-range wireless technology family.

5-Mark Answer

Explain the basic characteristics and use cases of 802.11 and Bluetooth.

10-Mark Answer

Compare 802.11 WLAN and Bluetooth in terms of range, network role, medium access and applications.

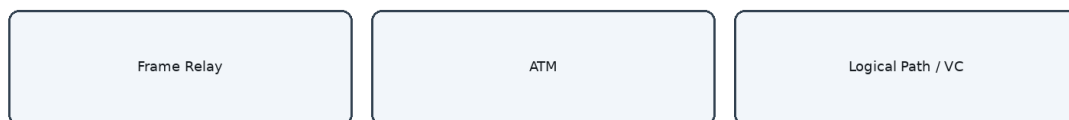
Chapter 12

Virtual-Circuit Networks: Frame Relay and ATM

Definition

Learning objectives: Explain virtual-circuit concepts and outline Frame Relay and ATM.

Virtual-Circuit Networks



12.1 Virtual Circuit

A virtual circuit is a logical path established through a packet-switched network. Packets/frames belonging to the connection follow the logical circuit and use an identifier rather than carrying a complete destination decision at every hop in the same manner as a pure datagram service.

12.2 Frame Relay

Frame Relay is a WAN technology using virtual circuits and frame switching, historically associated with efficient digital links and relatively low protocol overhead.

12.3 ATM

Asynchronous Transfer Mode uses fixed-size 53-byte cells: 5-byte header and 48-byte payload. It supports virtual paths/channels and was designed for integrated voice, video and data transport.

Important Exam Point

ATM's 53-byte cell is a common examination fact: 5-byte header + 48-byte payload.

2-Mark Answer

A virtual circuit is a logical communication path through a packet-switched network.

5-Mark Answer

Explain virtual circuits and briefly describe Frame Relay and ATM.

10-Mark Answer

Discuss virtual-circuit networks, Frame Relay and ATM, including ATM cell structure and applications.

Part III

UNIT III – NETWORK LAYER

Chapter 13

Logical Addressing: IPv4 and IPv6 Addresses

Definition

Learning objectives: Explain logical addressing, IPv4 structure, IPv6 structure and basic notation.

IPv4 / IPv6 Concept



13.1 Logical Addressing

Network-layer logical addresses identify interfaces/nodes for internetwork delivery. IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses.

13.2 IPv4

IPv4 addresses are written in dotted-decimal notation, e.g. 192.0.2.10. Prefix notation such as /24 identifies the network-prefix length.

13.3 IPv6

IPv6 addresses are 128 bits and commonly written in hexadecimal groups separated by colons. Leading zeros within a group may be suppressed; one contiguous run of zero groups may be compressed using

::.

13.4 Addressing Concepts

Network prefix, interface/host portion, subnetting and prefix aggregation are central ideas. Private IPv4 address space and IPv6 address scopes are distinct concepts and should not be conflated.

Numerical Example

For an IPv4 /24 prefix, 24 bits belong to the prefix and 8 bits remain in the host/interface portion. Under traditional subnet calculations, the number of addresses is $2^8 = 256$; usable-host calculations depend on the addressing context and special-purpose addresses.

2-Mark Answer

IPv4 uses 32-bit addresses; IPv6 uses 128-bit addresses.

5-Mark Answer

Explain IPv4 notation and IPv6 notation with examples.

10-Mark Answer

Explain logical addressing, IPv4/IPv6 formats, prefix notation and subnetting concepts.

Chapter 14

Internet Protocol: IPv4 and IPv6

Definition

Learning objectives: Explain the network-layer forwarding role of IPv4/IPv6 and major differences.

14.1 IP Function

IP provides best-effort network-layer datagram delivery across interconnected networks. Routers forward packets according to destination address and routing information.

14.2 IPv4

IPv4 includes a variable-length header with fields such as source/destination addresses, TTL, protocol and checksum. Routers can fragment IPv4 packets under the IPv4 rules, although modern networks often use path-MTU discovery to avoid fragmentation.

14.3 IPv6

IPv6 has a fixed 40-byte base header and extension headers. It replaces TTL with Hop Limit and uses Next Header to identify an upper-layer protocol or extension header. IPv6 routers do not fragment packets; fragmentation is performed by the source using an extension header when needed.

14.4 Comparison

	IPv4	IPv6
Address length	32 bits	128 bits
Base header	Variable, minimum 20 bytes	Fixed 40 bytes
TTL equivalent	TTL	Hop Limit
Router fragmentation	Permitted under IPv4 rules	Not performed by routers
Header checksum	Present in IPv4 header	No base-header checksum

2-Mark Answer

IP provides best-effort datagram delivery using logical addresses and routing.

5-Mark Answer

List key IPv4 and IPv6 differences.

10-Mark Answer

Explain IPv4 and IPv6 operation and compare addressing, headers, fragmentation and control fields.

Chapter 15

Transition from IPv4 to IPv6: Dual Stack and Tunneling

Definition

Learning objectives: Explain why transition mechanisms are needed and describe dual stack and tunneling.

15.1 Need for Transition

IPv4 and IPv6 are different network protocols. During migration, networks may need mechanisms allowing IPv4 and IPv6 systems to coexist.

15.2 Dual Stack

A dual-stack host/network supports both IPv4 and IPv6 and selects the appropriate protocol for communication.

15.3 Tunneling

Tunneling carries one protocol inside another across a portion of the path. For example, IPv6 packets can be encapsulated inside IPv4 to cross an IPv4-only segment, subject to the particular tunnel mechanism.

Important Exam Point

Dual stack uses both protocols directly; tunneling transports one protocol through another protocol's infrastructure.

2-Mark Answer

Dual stack means supporting IPv4 and IPv6 simultaneously.

5-Mark Answer

Explain dual stack and tunneling.

10-Mark Answer

Discuss IPv4-to-IPv6 transition, need, dual stack, tunneling and their advantages/limitations.

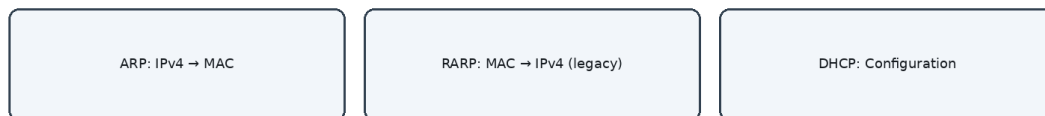
Chapter 16

Address Mapping: ARP, RARP and DHCP

Definition

Learning objectives: Explain address mapping/configuration functions of ARP, legacy RARP and DHCP.

Address Mapping



16.1 ARP

Address Resolution Protocol maps an IPv4 protocol address to a link-layer address on the local network. A host may broadcast an ARP request and the target can reply with its link-layer address.

16.2 RARP

Reverse ARP historically allowed a host to obtain an IPv4 address using a link-layer address. It has been superseded by more capable configuration mechanisms such as BOOTP and DHCP.

16.3 DHCP

DHCP dynamically supplies configuration information such as an IP address, subnet parameters and gateway/DNS information according to the deployment.

DHCP Basic Exchange**2-Mark Answer**

ARP resolves an IPv4 address to a link-layer address on the local network.

5-Mark Answer

Differentiate ARP, RARP and DHCP.

10-Mark Answer

Explain ARP operation, the historical role of RARP and the DHCP address/configuration process.

Chapter 17

Network Layer Protocols: ICMP and IGMP

Definition

Learning objectives: Explain the purposes of ICMP and IGMP and distinguish control messaging from data forwarding.

Control / Group Protocols



17.1 ICMP

Internet Control Message Protocol supports control, diagnostic and error-reporting messages associated with IP. Examples include destination-unreachable and time-exceeded messages; ping uses ICMP Echo messages.

17.2 IGMP

Internet Group Management Protocol supports IPv4 multicast group membership management between hosts and local multicast routers.

Common Mistake

ICMP is not a replacement for TCP/UDP application data. It provides IP-related control and diagnostic messaging.

2-Mark Answer

ICMP provides IP control/diagnostic messaging; IGMP manages IPv4 multicast group membership.

5-Mark Answer

Explain ICMP and IGMP with examples.

10-Mark Answer

Discuss the purpose, major message categories and applications of ICMP and IGMP.

Chapter 18

Routing: Unicast, RIP, OSPF, Distance Vector and Link State

Definition

Learning objectives: Explain routing, unicast forwarding, distance-vector and link-state concepts, RIP and OSPF.

Routing Concepts



18.1 Routing

Routing determines paths through an internetwork so packets can reach destination networks.

18.2 Unicast Routing

One sender communicates with one intended destination. Routers use forwarding tables populated by static configuration or routing protocols.

18.3 Distance Vector

A router exchanges reachability/distance information with neighbors. The Bellman-Ford family of ideas underlies classic distance-vector routing. RIP is a classic example and uses hop count as its metric, with a maximum usable metric of 15 and 16 representing unreachable.

18.4 Link State

Routers build a topology view from link-state information and run a shortest-path computation. OSPF is a major link-state interior gateway protocol and uses a cost metric.

18.5 Comparison

	Distance Vector	Link State
Knowledge	Neighbor-derived route information	More complete topology information
Algorithm family	Bellman-Ford style	Dijkstra shortest-path style
Example	RIP	OSPF
Convergence	Can be slower in classic designs	Typically faster in well-designed link-state domains

2-Mark Answer

Routing selects paths for forwarding packets toward destination networks.

5-Mark Answer

Differentiate distance-vector and link-state routing.

10-Mark Answer

Explain unicast routing, distance vector, link state, RIP and OSPF with a comparison.

Chapter 19

Multicast Routing: DVMRP and MOSPF

Definition

Learning objectives: Explain multicast routing and outline DVMRP and MOSPF at the syllabus level.

19.1 Multicast

Multicast delivers traffic from one source to a group of receivers. It can reduce duplicate traffic compared with separate unicast streams when many receivers need the same content.

19.2 DVMRP

Distance Vector Multicast Routing Protocol is a multicast routing protocol based on distance-vector ideas and techniques such as reverse-path forwarding/pruning.

19.3 MOSPF

Multicast OSPF extends OSPF concepts for multicast routing by incorporating multicast information into an OSPF-based link-state environment.

Important Exam Point

For examinations, emphasize the contrast: DVMRP is distance-vector based; MOSPF extends link-state/OSPF concepts for multicast.

2-Mark Answer

Multicast routing delivers packets from one source to multiple members of a group.

5-Mark Answer

Explain DVMRP and MOSPF.

10-Mark Answer

Discuss multicast routing, its need, DVMRP operation at a conceptual level and MOSPF.

Part IV

UNIT IV – TRANSPORT LAYER AND SECURITY

Chapter 20

Lesson 1: Process-to-Process Delivery

Definition

Learning objectives: Understand the transport layer's role, process-to-process communication, port numbers, multiplexing and demultiplexing.

20.1 Introduction

The transport layer provides logical communication between application processes running on different hosts. The network layer delivers packets from host to host; the transport layer uses transport addresses, especially port numbers, to deliver data to the correct application endpoint. TCP and UDP are the principal Internet transport protocols discussed in this unit.

Process-to-Process Delivery



IP identifies hosts; transport-layer port numbers identify application endpoints/processes.

20.2 Process-to-Process Communication

A **process** is a running program instance. A host may run many network applications simultaneously. A transport protocol therefore needs to distinguish not only the destination host but also the destination process. A transport-layer endpoint is commonly identified using an IP address together with a port number and protocol.

- **Host-to-host delivery:** primarily the responsibility of the network layer.
- **Process-to-process delivery:** transport-layer responsibility.

- **Port number:** identifies the application service/process endpoint within the transport protocol's namespace.

20.3 Port Numbers

TCP and UDP use 16-bit source and destination port fields. Port numbers range from 0 to 65535. The port space is commonly described as:

- **Well-known ports:** 0–1023.
- **Registered ports:** 1024–49151.
- **Dynamic/private ports:** 49152–65535.

The ranges describe conventional registration/assignment categories; an application should follow the current IANA service-name/port-number registry rather than memorizing a service from a textbook table.

20.4 Multiplexing and Demultiplexing

Multiplexing occurs at the sender: data from multiple application processes is accepted by the transport layer, associated with transport headers, and passed to the network layer. **Demultiplexing** occurs at the receiver: the transport layer examines the transport header and delivers the payload to the appropriate receiving socket/process.

1. Application process writes data to its socket.
2. Transport protocol adds a header containing ports and other control information.
3. Network layer carries the resulting transport segment/datagram.
4. Receiver's transport layer examines the destination port and protocol state.
5. Payload is delivered to the appropriate socket and process.

Important Exam Point

Exam point: Multiplexing = many application flows → one transport service at the sender.
Demultiplexing = received transport data → the correct receiving application endpoint.

20.5 Real-World Example

A laptop may simultaneously use a browser, DNS resolver, video call and file-transfer application. Port numbers allow the host's networking stack to keep these flows logically separate even though they share the same network interface.

20.6 Advantages and Limitations

Advantages: process identification, concurrent application communication, protocol-specific reliability/flow control when TCP is used, and standardized application endpoints.

Limitations: port numbers alone do not provide security or application authentication; delivery still depends on the selected transport protocol and lower layers.

2-Mark Answer

Process-to-process delivery is delivery of data from an application process on one host to the corresponding application process on another host using transport-layer addressing.

5-Mark Answer

Define process-to-process communication, explain port numbers, and distinguish multiplexing from demultiplexing with a diagram.

10-Mark Answer

Explain the transport layer's process-to-process delivery function in detail. Cover processes, ports, multiplexing, demultiplexing, operation and an example.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 21

Lesson 2: UDP User Datagram Format

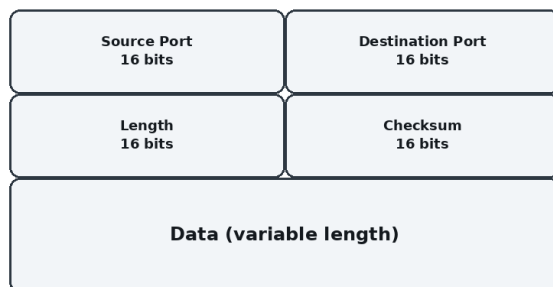
Definition

Learning objectives: Describe UDP, its fixed 8-byte header, every field, characteristics, advantages, limitations and applications.

21.1 UDP Overview

The User Datagram Protocol (UDP) is a connectionless, transaction-oriented transport protocol. RFC 768 specifies a minimal transport mechanism with source and destination ports, length and checksum. It does not provide TCP-style reliable, ordered byte-stream delivery or connection establishment. [5]

UDP User Datagram Format — 8-byte header



Length = header + data; minimum UDP length is 8 bytes.

21.2 UDP User Datagram Format

The UDP header is 8 bytes:

Field	Purpose	Size
Source Port	Port of the sending application endpoint; may be zero when not used	16 bits
Destination Port	Port of the receiving application endpoint	16 bits
Length	Total UDP datagram length, including header and data	16 bits
Checksum	One's-complement checksum over the UDP pseudo-header, UDP header and data	16 bits

The minimum UDP length is 8 bytes because the header itself is 8 bytes. [5]

21.3 UDP Characteristics

- Connectionless: no TCP-style handshake before sending data.
- Message/datagram oriented: each UDP send corresponds to a datagram boundary.
- No built-in retransmission, sequencing for ordered delivery, receiver flow control, or congestion control equivalent to TCP's mechanisms.
- Small fixed header and low protocol overhead.
- Supports port-based multiplexing/demultiplexing.

21.4 Advantages

Low overhead, no connection-establishment delay, and useful for applications that can tolerate loss or implement their own reliability/timing mechanisms.

21.5 Limitations

UDP itself does not guarantee delivery, ordering, duplicate suppression or congestion control. An application requiring such properties must implement them itself or use a protocol/framework that supplies them.

21.6 Applications

UDP is used by many applications and protocols where low overhead, datagram semantics, or application-controlled behavior are useful, including DNS queries, DHCP, streaming/media protocols and modern transports such as QUIC (which uses UDP as its substrate).

Common Mistake

Do not write “UDP is unreliable and therefore useless.” The precise statement is that UDP does not provide TCP-style reliability and ordering; applications can add reliability or other controls above UDP.

2-Mark Answer

UDP is a connectionless transport protocol with an 8-byte header containing source port, destination port, length and checksum.

5-Mark Answer

Draw and explain the UDP datagram format and list its major characteristics.

10-Mark Answer

Explain UDP in detail: purpose, header format, field functions, operations/characteristics, advantages, limitations and applications.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 22

Lesson 3: UDP Operations and Checksum

Definition

Learning objectives: Explain UDP send/receive operations and solve one's-complement checksum problems accurately.

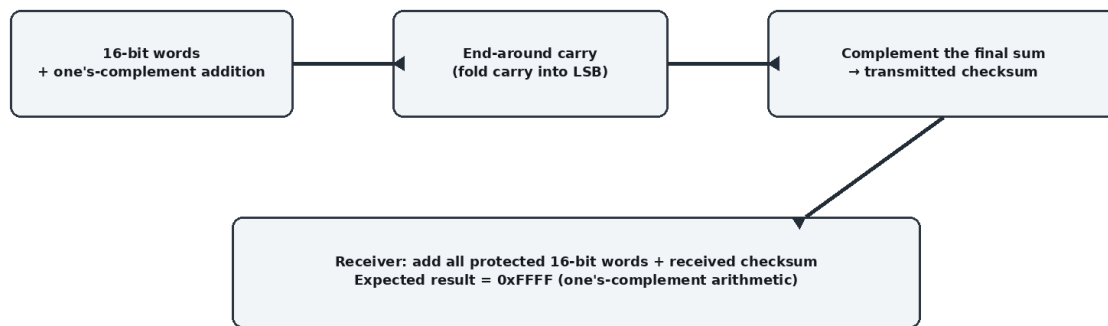
22.1 UDP Operations

UDP communication is simple:

1. The application supplies a message to UDP.
2. UDP identifies the source and destination ports.
3. UDP forms a datagram and calculates the checksum when checksum protection is used.
4. The datagram is handed to IP for transmission.
5. At the receiver, UDP validates the checksum as applicable and uses the destination port to deliver the payload to the correct socket.

22.2 UDP Checksum

RFC 768 defines a 16-bit one's-complement checksum. The checksum calculation conceptually covers a pseudo-header derived from IP information, the UDP header, and the data. The pseudo-header protects against delivery to the wrong IP destination/protocol context. [5]

UDP Checksum – sender and receiver verification**22.2.1 One's-Complement Addition**

For a 16-bit one's-complement sum:

1. Add the 16-bit words using ordinary binary addition.
2. If a carry leaves the most-significant bit, add that carry back into the least-significant bit (end-around carry).
3. Continue folding carries if necessary.
4. Complement every bit of the final 16-bit sum to obtain the checksum.

Important Exam Point

For a receiver-side textbook verification problem, adding all protected words plus the received checksum should produce **FFFF** in one's-complement arithmetic when no error is detected.

22.3 Worked Checksum Problem 1

Given 16-bit words: 1234, 5678, 9ABC.

Step 1: Add

$$1234 + 5678 = 68AC$$

$$68AC + 9ABC = 10368$$

Step 2: End-around carry The result has a carry of 1 beyond 16 bits:

$$0368 + 1 = 0369$$

Step 3: Complement

$$\sim 0369 = FC96$$

Therefore, the transmitted checksum is **FC96**.

Step 4: Receiver verification

$$0369 + FC96 = FFFF$$

Thus the receiver's one's-complement sum is FFFF: the example passes the checksum test.

22.4 Worked Checksum Problem 2

Given 1111, 2222, 3333, 4444:

$$1111 + 2222 = 3333, \quad 3333 + 3333 = 6666, \quad 6666 + 4444 = AAAA.$$

There is no overflow carry. Complement:

$$\sim AAAA = 5555.$$

Receiver:

$$AAAA + 5555 = FFFF.$$

22.5 Worked Checksum Problem 3 — Carry

Given ABCD, 1234, FFFF:

$$ABCD + 1234 = BE01, \quad BE01 + FFFF = 1BE00.$$

Fold the carry:

$$BE00 + 1 = BE01.$$

Complement:

$$\sim BE01 = 41FE.$$

Receiver:

$$BE01 + 41FE = FFFF.$$

22.6 Important IPv4/IPv6 Note

RFC 768 describes the UDP checksum and permits a transmitted zero checksum value to mean that the sender generated no checksum in the applicable IPv4 context. IPv6 has a different rule: the UDP checksum is not optional by default; RFC 8200 requires it for UDP over IPv6 except for defined zero-checksum tunnel exceptions. [5, 7]

Common Mistake

Do not confuse UDP checksum with parity or CRC. UDP checksum uses one's-complement arithmetic over 16-bit words and includes the UDP pseudo-header in its calculation.

2-Mark Answer

UDP checksum is a 16-bit one's-complement checksum used to detect errors in the UDP pseudo-header, UDP header and data.

5-Mark Answer

Explain the UDP checksum algorithm and demonstrate one worked example.

10-Mark Answer

Explain UDP operations and checksum step by step. Include one's-complement addition, end-around carry and one numerical example.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 23

Lesson 4: TCP Services

Definition

Learning objectives: Explain TCP's connection-oriented service, reliability, byte-stream model, full-duplex communication, flow control and congestion-related behavior.

23.1 TCP Overview

The Transmission Control Protocol (TCP) is an Internet transport protocol that provides a reliable, in-order byte-stream service to applications. RFC 9293 is the current consolidated base TCP specification and replaces RFC 793 as the core specification. [6]

23.2 Major TCP Services

1. **Connection-oriented service:** endpoints establish a logical connection before normal data transfer.
2. **Reliable delivery:** sequence numbers, acknowledgments, checksums and retransmission mechanisms support reliable delivery.
3. **Ordered byte stream:** the application sees a continuous byte stream rather than independent message boundaries.
4. **Full-duplex transfer:** both directions can carry data simultaneously.
5. **Flow control:** the receiver advertises how much data it can accept.
6. **Congestion control:** the sender limits outstanding data according to network conditions.
7. **Multiplexing:** port numbers identify application services and help distinguish flows.

Important Exam Point

TCP reliability is not magic: it is built from mechanisms such as sequence numbers, acknowledgments, checksums and retransmission. TCP also uses receiver flow control and congestion control, which solve different problems.

23.3 TCP Connection States

Common states include LISTEN, SYN-SENT, SYN-RECEIVED, ESTABLISHED, FIN-WAIT-1, FIN-WAIT-2, CLOSE-WAIT, LAST-ACK, TIME-WAIT and CLOSED. The state machine is important for understanding connection setup and termination.

23.4 Applications

TCP is widely used for applications that need a reliable ordered byte stream, such as traditional web traffic, file transfer, remote login and many application protocols.

2-Mark Answer

TCP is a connection-oriented transport protocol providing a reliable, in-order byte-stream service.

5-Mark Answer

List and briefly explain the major services provided by TCP.

10-Mark Answer

Explain TCP services in detail and show why connection orientation, reliability, byte-stream delivery, flow control and congestion control are important.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 24

Lesson 5: TCP Segment Format

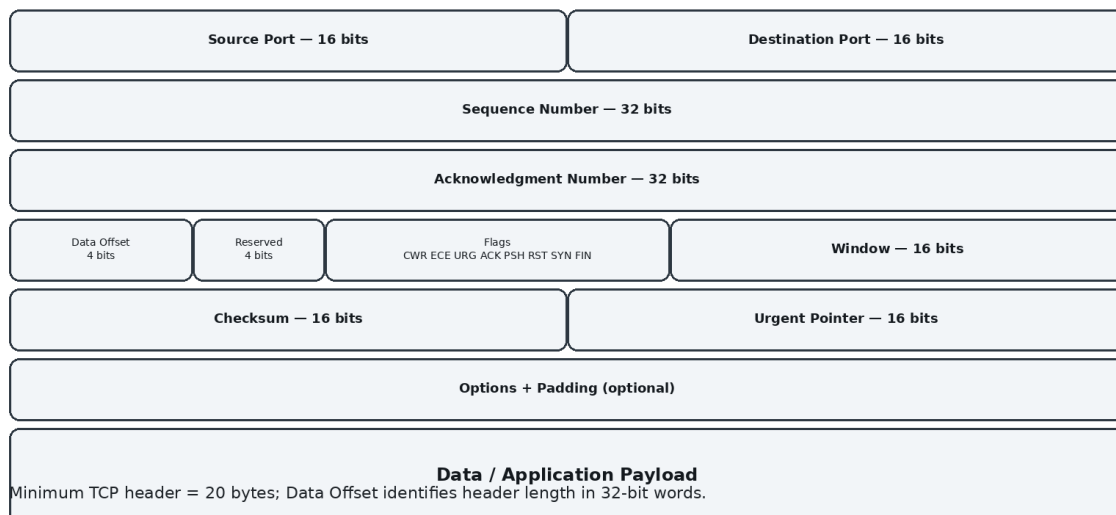
Definition

Learning objectives: Identify TCP header fields, their sizes and functions, including current TCP control flags.

24.1 TCP Segment

TCP carries application bytes in segments. Each segment has a TCP header followed by optional data. RFC 9293 specifies the header format. [6]

TCP Segment Format (RFC 9293 field structure)



24.2 TCP Header Fields

Field	Size	Purpose
Source Port	16 bits	Sending application endpoint
Destination Port	16 bits	Receiving application endpoint
Sequence Number	32 bits	Byte position of first data octet; special handling for SYN
Acknowledgment Number	32 bits	Next sequence number expected when ACK is set
Data Offset	4 bits	TCP header length in 32-bit words
Reserved	4 bits	Reserved for future use
Control flags	8 bits	CWR, ECE, URG, ACK, PSH, RST, SYN, FIN
Window	16 bits	Advertised receive window
Checksum	16 bits	Error detection over TCP segment and pseudo-header
Urgent Pointer	16 bits	Used with URG for urgent-data semantics
Options	Variable	Optional extensions, aligned with padding
Data	Variable	Application payload

24.3 Important Flags

- **SYN:** synchronize sequence numbers / initiate connection.
- **ACK:** acknowledgment field is valid.
- **FIN:** sender has no more data to send and requests orderly close.
- **RST:** reset/abort a connection or reject an inappropriate segment.
- **PSH:** requests prompt delivery of buffered data to the receiving application in the applicable TCP semantics.
- **URG:** urgent pointer is significant.
- **ECE/CWR:** related to Explicit Congestion Notification signaling.

Common Mistake

Older textbooks sometimes draw only six familiar flags (URG, ACK, PSH, RST, SYN, FIN). The current RFC 9293 header format also includes ECE and CWR. For an up-to-date answer, mention all eight while noting the six-flag legacy diagram if your university notes use it. [6]

2-Mark Answer

The TCP Data Offset field gives the length of the TCP header in 32-bit words.

5-Mark Answer

Draw the TCP segment format and explain the major fields.

10-Mark Answer

Draw and explain the TCP segment format, field sizes, flags and their functions. Mention the minimum header size and the role of options.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 25

Lesson 6: TCP Connection Establishment and Termination

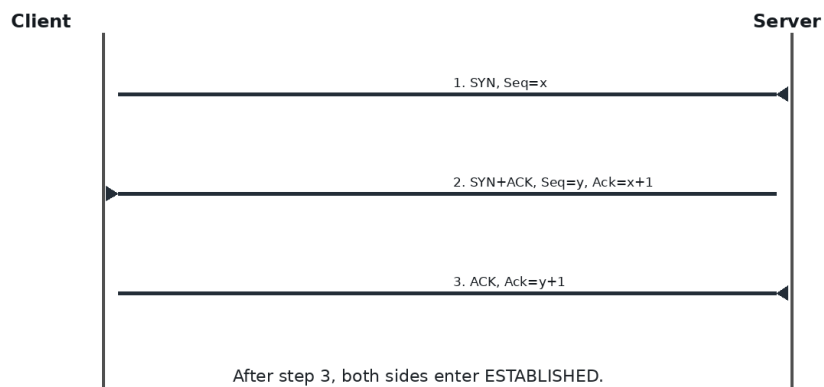
Definition

Learning objectives: Explain the three-way handshake, sequence-number synchronization and four-step orderly termination.

25.1 Three-Way Handshake

TCP uses a three-way exchange to establish a connection and synchronize initial sequence numbers.

TCP Three-Way Connection Establishment



1. **SYN:** client sends SYN with initial sequence number x .
2. **SYN+ACK:** server acknowledges x with ACK number $x + 1$ and sends its own initial sequence number y .
3. **ACK:** client acknowledges the server's SYN with ACK number $y + 1$.

After the third exchange, both endpoints can enter the ESTABLISHED state and exchange application data.

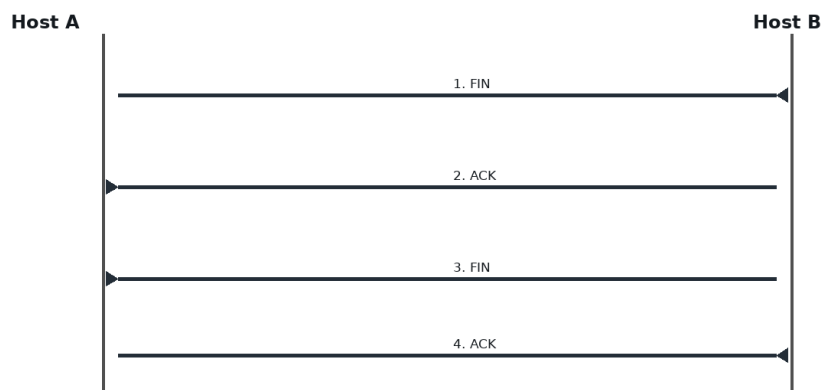
25.2 Why Three Messages?

The exchange lets each endpoint learn that the other endpoint can send and receive and establishes synchronized sequence-number state in both directions. It also avoids treating an old delayed SYN as sufficient evidence of a currently established two-way connection.

25.3 Connection Termination

A TCP connection is full-duplex, so each direction is closed independently. A common orderly close uses four control exchanges.

TCP Connection Termination — four control exchanges



1. Host A sends FIN.
2. Host B sends ACK for A's FIN.
3. Host B later sends its own FIN.
4. Host A sends the final ACK.

A FIN consumes one sequence number. TCP may enter TIME-WAIT after the active close so that delayed segments can expire and the final acknowledgment can be retransmitted if necessary.

25.4 Sequence-Number Example

Suppose a client sends SYN with sequence number 1000. The SYN consumes one sequence number, so the server's ACK is 1001. If the server sends SYN with sequence number 7000, the client's ACK is 7001.

Important Exam Point

Exam rule: SYN and FIN each consume one sequence number even though they do not carry application data. A pure ACK normally does not consume a sequence number.

2-Mark Answer

TCP connection establishment uses a three-way handshake: SYN, SYN-ACK and ACK.

5-Mark Answer

Explain TCP three-way handshake with a neat diagram and sequence/acknowledgment numbers.

10-Mark Answer

Explain TCP connection establishment and termination. Draw both diagrams and explain the role of SYN, ACK, FIN, sequence numbers, acknowledgment numbers and TIME-WAIT.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 26

Lesson 7: TCP Flow Control

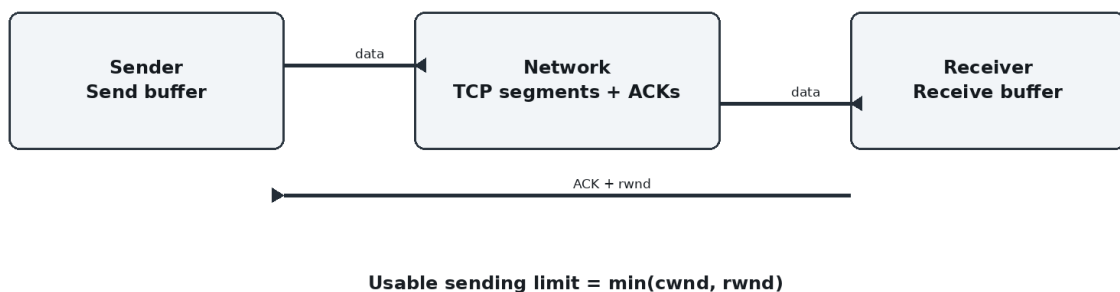
Definition

Learning objectives: Explain receiver-based flow control, advertised window, sliding window and the relationship between rwnd and cwnd.

26.1 Definition

Flow control prevents a fast sender from overwhelming a slower receiver. TCP implements receiver flow control using the advertised receive window (rwnd).

TCP Flow Control — receiver advertised window (rwnd)



26.2 How TCP Flow Control Works

1. The receiver has a finite receive buffer.
2. The receiver advertises available buffer space in the Window field.
3. The sender limits the amount of unacknowledged data so it does not exceed the receiver's advertised capacity.
4. As the application consumes received bytes, the receiver can advertise additional window space.

26.3 Sliding Window

TCP's sending window moves forward as acknowledgments arrive. Bytes below the left edge have been acknowledged; bytes inside the usable window may be in flight; bytes beyond the right edge cannot yet be sent.

26.4 Flow Control vs Congestion Control

Flow control protects the **receiver**. Congestion control protects the **network**. A TCP sender is constrained by both:

$$\text{sending limit} \approx \min(\text{rwnd}, \text{cwnd}).$$

This relationship is described in the TCP congestion-control standards. [10]

26.5 Numerical Example

Suppose:

$$\text{rwnd} = 4000 \text{ bytes}, \quad \text{cwnd} = 6000 \text{ bytes}.$$

Then the sender is limited to:

$$\min(4000, 6000) = 4000 \text{ bytes}$$

of outstanding data under this simplified model.

If the receiver later advertises $\text{rwnd} = 7000$ while cwnd remains 6000, the network-side congestion window becomes the limiting factor and the usable limit is 6000 bytes.

Common Mistake

Do not define flow control as “controlling traffic in the network.” That is congestion control. Flow control is specifically about matching sender transmission to receiver capacity.

2-Mark Answer

TCP flow control uses the receiver's advertised window to limit outstanding data.

5-Mark Answer

Explain TCP flow control and sliding window with an example.

10-Mark Answer

Explain TCP flow control in detail, including receive buffers, advertised window, sliding-window operation, numerical calculation and the difference from congestion control.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 27

Lesson 8: TCP Error Control

Definition

Learning objectives: Explain how TCP detects corruption and loss and recovers using checksum, sequence numbers, acknowledgments and retransmission.

27.1 Purpose

TCP error control aims to provide reliable byte-stream delivery despite corruption, loss, duplication or reordering.

TCP Error Control — sequence, ACK, checksum and retransmission



Checksum detects corruption; sequence numbers identify byte positions; ACKs report cumulative progress.

27.2 Main Mechanisms

1. **Checksum:** detects corruption in the TCP segment and protected pseudo-header information.
2. **Sequence numbers:** identify byte positions and permit the receiver to detect missing or duplicate data.
3. **Acknowledgments:** indicate cumulative reception progress.
4. **Retransmission timer:** if an expected acknowledgment does not arrive in time, the sender may retransmit.
5. **Duplicate ACKs / fast retransmit:** multiple duplicate acknowledgments can provide an early indication of loss.

6. **SACK and advanced loss recovery:** optional mechanisms can improve recovery from multiple losses.

27.3 Sequence and ACK Rule

If a segment starts with sequence number 1000 and carries 500 bytes, the next byte is 1500. A cumulative acknowledgment for all bytes through 1499 is:

$$ACK = 1500.$$

Thus the acknowledgment number identifies the **next byte expected**, not the last byte received.

27.4 Numerical Example

Segment 1: Seq = 5000, data length = 1000 bytes.
Therefore next expected byte = 6000.

$$ACK = 6000.$$

If the next segment begins at Seq = 6000 and carries 700 bytes:

$$ACK = 6700.$$

27.5 Loss Example

If the receiver receives segment beginning at 6000 before receiving the segment beginning at 5000, it cannot advance its cumulative acknowledgment past the missing byte range. The sender may receive duplicate ACKs indicating the receiver still expects the earlier sequence space. Under the classical fast-retransmit rule, three duplicate ACKs trigger retransmission without waiting for the retransmission timer. [10]

Common Mistake

Error control and congestion control are related but not identical. A retransmission repairs a delivery problem; congestion control decides how aggressively TCP should inject data into the network.

2-Mark Answer

TCP uses checksum, sequence numbers, acknowledgments and retransmission to support reliable delivery.

5-Mark Answer

Explain TCP error control with sequence numbers and acknowledgments.

10-Mark Answer

Explain TCP error control in detail, including checksum, sequence numbers, cumulative ACKs, timeout-based retransmission, duplicate ACKs and fast retransmit. Include a numerical example.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 28

Lesson 9: TCP Congestion Control

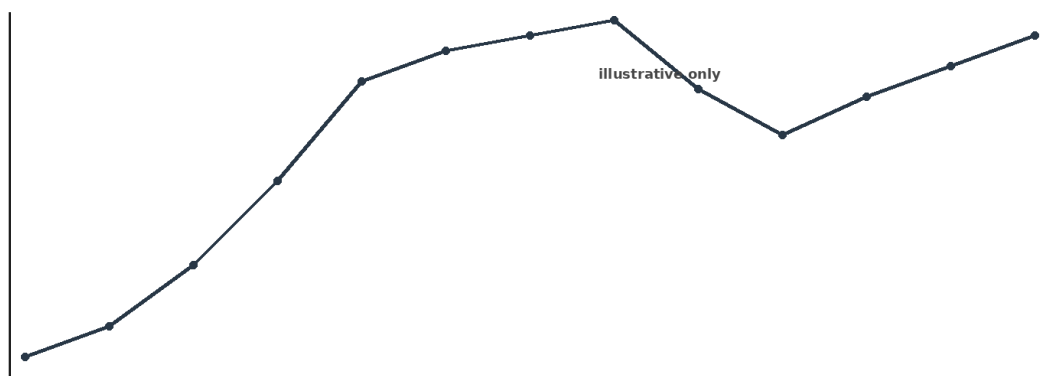
Definition

Learning objectives: Understand congestion, cwnd, ssthresh, slow start, congestion avoidance, fast retransmit and fast recovery.

28.1 Definition

Congestion occurs when offered traffic exceeds the capacity available along a path, causing queue buildup, delay and packet loss. TCP congestion control regulates the sender's amount of outstanding data to reduce the risk of overwhelming the network.

Conceptual TCP Congestion Window (cwnd)



Modern TCP variants differ in exact cwnd behavior; the graph illustrates the classical exam model.

28.2 Key Variables

- **cwnd:** sender-side congestion window; limits data injected into the network.
- **rwnd:** receiver-advertised window; limits data according to receiver capacity.
- **ssthresh:** threshold used by the classical algorithm to distinguish slow start from congestion avoidance.

The usable outstanding-data limit is governed by the smaller of cwnd and rwnd. [10]

28.3 Classical Four Algorithms

RFC 5681 describes four intertwined algorithms: slow start, congestion avoidance, fast retransmit and fast recovery. [10]

28.3.1 Slow Start

TCP begins by probing network capacity. In the classical educational model, $cwnd$ grows rapidly, approximately doubling each RTT while ACKs arrive, until $ssthresh$ or congestion is reached. Modern TCP initial-window behavior is more nuanced, so the exact numerical growth should be treated as a conceptual exam model unless a particular TCP variant is specified.

28.3.2 Congestion Avoidance

After the threshold, classical congestion avoidance grows $cwnd$ much more slowly, approximately one MSS per RTT. This is the additive-increase portion of the familiar AIMD model.

28.3.3 Fast Retransmit

Three duplicate ACKs are traditionally used as an indication of a missing segment, allowing retransmission before the retransmission timer expires. [10]

28.3.4 Fast Recovery

After fast retransmit, classical fast recovery reduces the sending rate while attempting to avoid returning immediately to the smallest possible window. Exact behavior depends on the TCP implementation and later loss-recovery standards.

28.4 Simple Numerical Example

Assume a textbook model:

$$cwnd = 1, 2, 4, 8$$

during slow start, with $ssthresh = 8$. Then congestion avoidance begins:

$$9, 10, 11, \dots$$

If a loss is detected by a timeout, the sender reduces its congestion window substantially and restarts growth according to the applicable TCP algorithm. If loss is detected by duplicate ACKs, fast retransmit/fast recovery may be used.

Important Exam Point

The classical graph is useful for exams, but do not claim that every modern TCP implementation follows exactly the same $cwnd$ curve. TCP congestion control has evolved through multiple RFCs and implementations. [6, 10]

2-Mark Answer

Congestion control regulates TCP sending behavior to reduce overload in the network; $cwnd$ is a central sender-side variable.

5-Mark Answer

Explain slow start and congestion avoidance with a cwnd graph.

10-Mark Answer

Explain TCP congestion control comprehensively: congestion, cwnd/rwnd/ssthresh, slow start, congestion avoidance, fast retransmit, fast recovery and a numerical illustration.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 29

Lesson 10: Quality of Service and Techniques

Definition

Learning objectives: Define QoS, explain bandwidth, delay, jitter, packet loss and reliability, and describe common techniques for improving QoS.

29.1 Definition

Quality of Service (QoS) describes the performance characteristics experienced by an application and the mechanisms used to manage those characteristics. Different applications require different service levels.

Quality of Service (QoS) Parameters



QoS aims to provide application-appropriate performance, especially for real-time traffic.

29.2 QoS Parameters

Parameter	Meaning	Example impact
Bandwidth	Data-carrying capacity available to a flow	Higher capacity can support higher-rate media
Delay	Time taken for data to travel end-to-end	High delay affects interactive applications
Jitter	Variation in packet delay	High jitter can disturb real-time audio/video
Packet Loss	Fraction of packets not successfully delivered	Loss may cause gaps, retransmissions or quality reduction
Reliability	Consistency/correctness of delivery	Important for transactions and critical data

29.3 Techniques to Improve QoS

1. **Overprovisioning:** provide more capacity than average demand.
2. **Traffic shaping:** regulate the rate or burstiness of traffic, e.g., leaky-bucket or token-bucket concepts.
3. **Scheduling/queuing:** prioritize or fairly schedule packets using appropriate queue disciplines.
4. **Admission control:** reject or defer new flows when resources cannot support required service.
5. **Resource reservation:** reserve network resources where the architecture supports it.
6. **Traffic classification and marking:** classify traffic so downstream devices can apply appropriate forwarding treatment.
7. **Congestion management/avoidance:** reduce overload and queue buildup.

29.4 Real-World Example

Interactive voice/video is sensitive to delay and jitter, while bulk file transfer is usually more tolerant of delay but needs reliable delivery. Therefore a network may use different traffic-management policies for different classes.

Common Mistake

QoS does not mean “maximum speed for every application.” It means managing measurable performance characteristics according to application and policy requirements.

2-Mark Answer

QoS is the set of performance characteristics and mechanisms used to provide appropriate network service, commonly considering bandwidth, delay, jitter and loss.

5-Mark Answer

Define QoS and explain its major parameters.

10-Mark Answer

Explain QoS in detail and discuss bandwidth, delay, jitter, packet loss, reliability and techniques such as traffic shaping, scheduling, admission control and resource reservation.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 30

Lesson 11: Introduction to Network Security

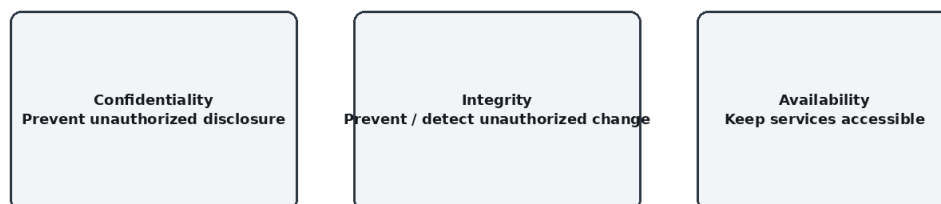
Definition

Learning objectives: Explain security objectives, threats, controls and the CIA triad in a networking context.

30.1 Definition

Network security protects networked information systems, communications and services against unauthorized access, disclosure, modification, disruption and misuse.

CIA Triad — Core Security Objectives



Security controls are selected according to the assets, threats and required objectives.

30.2 CIA Triad

- **Confidentiality:** information is not disclosed to unauthorized entities.
- **Integrity:** information is not changed in an unauthorized or undetected manner.
- **Availability:** authorized users can access systems and information when required.

30.3 Security Threats

Threats may include eavesdropping, tampering, spoofing, replay, malware, denial-of-service, credential theft, phishing and social engineering.

30.4 Security Services and Controls

Common security mechanisms include encryption, hashing/MACs, digital signatures, authentication, authorization, access control, secure protocols, logging, monitoring, backups and incident response.

30.5 Attack Categories

Passive attacks primarily observe information, such as eavesdropping. **Active attacks** attempt to alter, inject, replay or disrupt information or service.

Important Exam Point

A security control should be selected against a threat and a security objective. Encryption primarily addresses confidentiality; authenticated integrity mechanisms address modification detection; authentication establishes identity or control of an authenticator.

2-Mark Answer

The CIA triad consists of confidentiality, integrity and availability.

5-Mark Answer

Explain network security and the CIA triad with examples.

10-Mark Answer

Introduce network security, explain CIA objectives, common threats, security services and controls, and show how controls map to security goals.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 31

Lesson 12: Symmetric-Key and Asymmetric Cryptography

Definition

Learning objectives: Compare secret-key and public-key cryptography, explain keys, operation, strengths, limitations and typical uses.

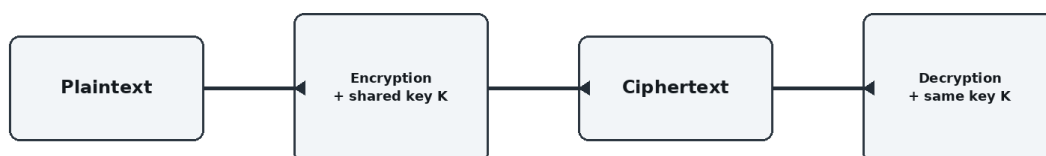
31.1 Cryptography

Cryptography uses algorithms and cryptographic keys to protect information. NIST guidance treats cryptographic mechanisms as tools for services such as confidentiality, integrity, authentication and digital signatures. [12]

31.2 Symmetric-Key Cryptography

In symmetric cryptography, communicating parties use the same secret key, or closely related secret key material, for encryption/decryption or other symmetric cryptographic operations.

Symmetric-Key Cryptography



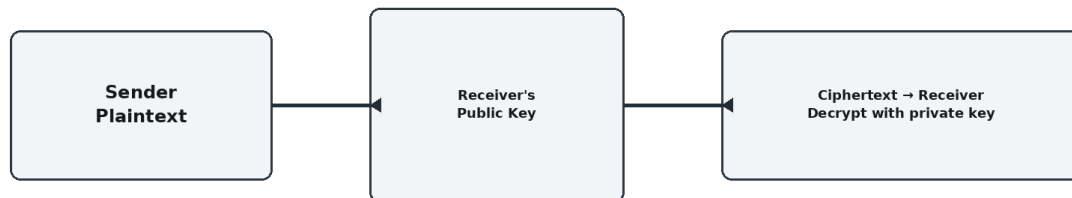
The communicating parties must possess the secret key securely.

Examples of standardized symmetric algorithms include AES-family encryption. The main practical challenge is secure key establishment and management.

31.3 Asymmetric Cryptography

Asymmetric cryptography uses a key pair: a public key that may be distributed and a corresponding private key that must be protected.

Asymmetric Cryptography — Public and Private Keys



Public key may be shared; private key must remain secret.

Asymmetric techniques support public-key encryption, key establishment and digital signatures. NIST cryptographic guidance covers asymmetric algorithms, key establishment, key management and digital signatures. [12]

31.4 Comparison

Feature	Symmetric	Asymmetric
Keys	Shared secret key(s)	Public/private key pair
Typical operation	Fast bulk cryptographic processing	Generally more computationally expensive; useful for key establishment/signatures
Key distribution	Shared secret must be established securely	Public key can be distributed; private key remains secret
Typical role	Bulk data protection	Key establishment, signatures, identity binding
Examples	AES	RSA, elliptic-curve public-key schemes

Common Mistake

Do not say “asymmetric encryption replaces symmetric encryption.” Secure systems commonly use asymmetric techniques to establish/authenticate keys and symmetric cryptography for efficient bulk protection.

2-Mark Answer

Symmetric cryptography uses shared secret key material; asymmetric cryptography uses a public/private key pair.

5-Mark Answer

Differentiate symmetric and asymmetric cryptography with examples and applications.

10-Mark Answer

Explain symmetric and asymmetric cryptography, their operation, key-management issues, advantages/limitations and a comparison table.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 32

Lesson 13: Message Security

Definition

Learning objectives: Explain confidentiality, integrity, message digests, MACs and encryption as message-security mechanisms.

32.1 Meaning

Message security protects a message against unauthorized disclosure, modification, forgery or misuse. Different mechanisms provide different services.

Message Security Services



32.2 Hash Functions

A cryptographic hash function maps an input of arbitrary length to a fixed-length digest. A secure hash is designed so that it is computationally difficult to find collisions and to recover the input from the digest under the appropriate security properties. NIST publications discuss hash functions and their use in digital signatures, HMAC and related applications. [13]

32.3 Message Authentication Code (MAC)

A MAC uses a shared secret key and a message to produce an authentication tag. A valid tag provides integrity/authentication evidence to parties that possess the shared secret, subject to the security of the MAC and key.

32.4 Encryption

Encryption transforms plaintext into ciphertext using an algorithm and key so that an authorized party can recover the protected information. Encryption is primarily a confidentiality mechanism; encryption alone should not be assumed to provide strong integrity/authentication.

32.5 Comparison: Encryption vs Hashing

Property	Encryption	Cryptographic Hash
Goal	Confidentiality	Digest/integrity-related applications
Reversible?	Designed to be decryptable with the appropriate key	Not designed to be reversed
Output	Ciphertext, generally related to input length	Fixed-length digest for a given hash function
Key required?	Yes	No for ordinary hash function
Common use	Confidential data	Integrity checks, signatures, password-verifier designs, KDF/MAC constructions as specified

Important Exam Point

A digital signature normally signs a digest or an equivalent standardized representation of the message; it is not the same thing as encrypting the message.

2-Mark Answer

A cryptographic hash produces a fixed-length digest; a MAC uses a secret key to authenticate a message.

5-Mark Answer

Explain message security using hashing, MAC and encryption.

10-Mark Answer

Discuss message security services and mechanisms. Explain hash functions, MACs and encryption, and compare encryption with hashing.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 33

Lesson 14: Digital Signature

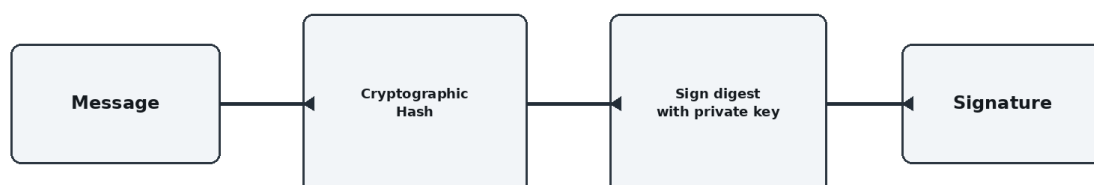
Definition

Learning objectives: Explain signature generation and verification, security properties, key roles, and the distinction between signatures and encryption.

33.1 Definition

A **digital signature** is a cryptographic mechanism that allows a verifier to detect unauthorized modification and authenticate the identity associated with the signing key, when the key is properly bound to that identity. NIST FIPS 186-5 specifies algorithms for generating digital signatures and describes their use for integrity and signer authentication. [8]

Digital Signature — Sign with Private Key, Verify with Public Key



Receiver recomputes the message digest and verifies the signature using the signer's public key.

33.2 Signature Generation

A simplified conceptual process is:

1. Compute a cryptographic digest of the message.
2. Use the signer's private key and the standardized signature algorithm to generate the signature value.
3. Send the message and signature.

33.3 Signature Verification

1. Receiver obtains the message and signature.
2. Receiver computes the required digest/representation of the received message.
3. Receiver uses the signer's public key and the signature-verification algorithm.
4. If verification succeeds, the signature is consistent with the message and public key.

33.4 Security Properties

Digital signatures can provide message integrity and signer authentication and can support non-repudiation claims in appropriate systems. The strength of identity binding depends on key management and the mechanism used to establish trust in the public key. [8]

33.5 Digital Signature vs Encryption

	Digital Signature	Encryption
Primary goal	Integrity + signer authentication	Confidentiality
Private/public key role	Sign with private key; verify with public key	Depends on scheme; e.g. encrypt to recipient's public key and decrypt with private key
Does it hide message?	No, not by itself	Yes, when correctly implemented
Trust requirement	Authentic public key binding is essential	Recipient public-key authenticity is also important

Common Mistake

A signature does not automatically make a message confidential. Signing and encryption are separate security services and can be combined when both are needed.

2-Mark Answer

A digital signature is a cryptographic mechanism used to detect unauthorized modification and authenticate the signatory.

5-Mark Answer

Explain digital signature generation and verification.

10-Mark Answer

Explain digital signatures in detail, including hash/digest, private-key signing, public-key verification, security properties, trust/key binding and the difference from encryption.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 34

Lesson 15: User Authentication

Definition

Learning objectives: Explain authentication, authenticators, factors, verification and the difference between authentication and authorization.

34.1 Definition

Authentication is the process of establishing that a claimant controls or possesses an authenticator associated with an identity. NIST's current Digital Identity Guidelines, SP 800-63B-4, define requirements for digital authentication and authenticator management. [9]

User Authentication



34.2 Authentication Factors

A common educational classification is:

- **Something you know:** password, PIN or other memorized secret.
- **Something you have:** a hardware token, authenticator device or cryptographic credential.
- **Something you are:** a biometric characteristic.

Multi-factor authentication combines independent factor categories rather than merely requiring two pieces of the same factor.

34.3 Authentication Process

1. User/claimant identifies the account or identity.
2. Claimant presents an authenticator.
3. A verifier checks the authenticator using the appropriate protocol.
4. The system determines whether the authentication succeeds.
5. A successful result can be used by an authorization system to decide what actions are permitted.

34.4 Authentication vs Authorization

	Authentication	Authorization
Question	Who/what is the claimant?	What is the authenticated entity allowed to do?
Timing	Establishes identity/control	Evaluates permissions
Example	Password + second factor	Permission to read a file

Common Mistake

Authentication is not authorization. A user can be successfully authenticated and still lack permission to perform a particular operation.

2-Mark Answer

Authentication establishes that a claimant controls an authenticator associated with an identity.

5-Mark Answer

Explain authentication factors and the authentication process.

10-Mark Answer

Explain user authentication, authentication factors, verifier operation, multi-factor authentication and authentication versus authorization.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 35

Lesson 16: E-mail Security

Definition

Learning objectives: Explain e-mail threats and the roles of S/MIME, OpenPGP concepts, encryption and digital signatures.

35.1 Need for E-mail Security

Ordinary e-mail is exposed to threats involving interception, modification, impersonation, malicious attachments, phishing and credential theft. Security mechanisms should address confidentiality, integrity and authentication as required.

E-mail Security — S/MIME / OpenPGP Concepts



Digital signatures support authentication/integrity; encryption supports confidentiality.

35.2 S/MIME

S/MIME is a standard for secure MIME data. RFC 8551 describes S/MIME 4.0 and its cryptographic services, including authentication, message integrity and non-repudiation through digital signatures, and confidentiality through encryption. [11]

35.3 OpenPGP Concept

OpenPGP provides mechanisms for protecting messages using public-key and symmetric cryptography, digital signatures and key management. In an exam answer, emphasize the service model rather than

assuming that a single key is used for every message.

35.4 Typical Secure E-mail Operation

1. Sender prepares the message.
2. For authenticity/integrity, the sender may generate a digital signature.
3. For confidentiality, the message may be encrypted for the intended recipient.
4. The secure message travels through ordinary mail infrastructure.
5. Receiver verifies the signature and/or decrypts the content using the required keys.

35.5 Security Services

Service	Typical mechanism
Confidentiality	Message encryption
Integrity	Signature/MAC and secure processing
Authentication	Digital signature and trusted key association
Non-repudiation support	Digital signatures in appropriate legal/technical contexts

Common Mistake

E-mail security does not eliminate phishing or social engineering. A cryptographically valid message can still be sent by a compromised or malicious account; users must evaluate context and trust.

2-Mark Answer

S/MIME provides secure MIME messaging services including authentication, integrity, digital signatures and encryption.

5-Mark Answer

Explain e-mail security and the role of S/MIME.

10-Mark Answer

Explain e-mail security in detail, covering threats, confidentiality, integrity, authentication, digital signatures, encryption, S/MIME and OpenPGP concepts.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 36

Lesson 17: Web Security and Social Issues

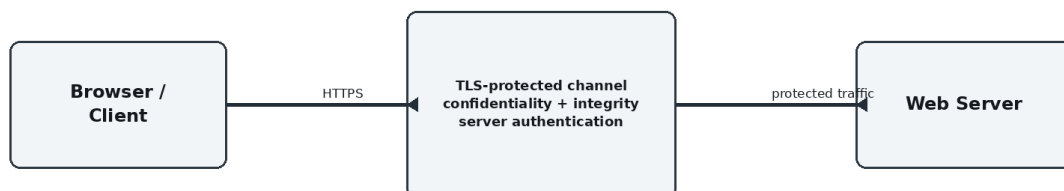
Definition

Learning objectives: Explain HTTPS/TLS at a conceptual level and discuss major social, ethical and human factors in network security.

36.1 Web Security

Web security protects browser-server communication, user accounts, application data and web services.

Web Security — HTTPS/TLS Concept



TLS secures application traffic; HTTPS is HTTP carried over TLS.

36.2 HTTPS and TLS

HTTPS is HTTP carried over a TLS-protected connection. TLS provides cryptographic mechanisms for confidentiality and integrity and can authenticate the server through certificates and the public-key infrastructure used by the deployment. The exact TLS version and cipher suites are protocol-specific.

36.3 Common Web Threats

- Credential theft and phishing.
- Session theft and insecure session management.

- Injection attacks, including SQL injection.
- Cross-site scripting (XSS).
- Cross-site request forgery (CSRF).
- Malicious downloads and compromised dependencies.
- Misconfiguration and insecure access control.

These are application-security concerns; TLS cannot fix vulnerabilities inside an application after the traffic reaches the server.

36.4 Security Practices

Use current secure protocol versions, protect credentials, validate and encode untrusted input appropriately, enforce authorization on the server, manage sessions securely, protect secrets and keys, apply security updates, log important security events and minimize unnecessary exposure.

36.5 Social Issues

Social Issues in Network Security



Technical controls are only one part of security; human, ethical, legal and societal factors also matter.

Network security has social and ethical dimensions:

- **Privacy:** collection and misuse of personal information.
- **Cybercrime:** malicious activity that harms systems, users or organizations.
- **Intellectual property:** unauthorized copying or distribution of digital works.
- **Digital divide:** unequal access to digital infrastructure and skills.
- **Ethics and responsible use:** security testing and research should respect authorization, safety and applicable rules.
- **Social engineering:** attackers exploit human trust, urgency or fear rather than only technical weaknesses.

Important Exam Point

Security is socio-technical: strong algorithms cannot compensate for stolen credentials, unsafe configurations, poor access control or manipulated users.

2-Mark Answer

HTTPS is HTTP carried over TLS; TLS provides cryptographic protection for the connection.

5-Mark Answer

Explain web security and list major social issues related to computer networks.

10-Mark Answer

Explain HTTPS/TLS, common web threats and security practices, then discuss privacy, cybercrime, intellectual property, digital divide, ethics and social engineering.

Exam Answer Framework

For a 10-mark answer: write the introduction/definition, explain the mechanism or format, draw the relevant diagram, include an example or numerical calculation where applicable, state key advantages/limitations or applications, and conclude with the central point.

Chapter 37

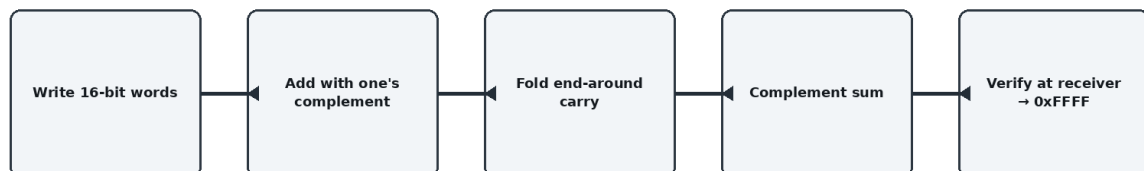
Lesson 18: Numerical Problems and Complete Unit Revision

Definition

Learning objectives: Practice checksum, parity, TCP sequence/ACK, window and congestion calculations and consolidate the complete unit.

37.1 A. UDP Checksum Practice

Checksum Worked-Problem Flow



37.1.1 Problem 1

Given 16-bit words 1234, 5678 and 9ABC. Find the checksum.

Solution:

$$1234 + 5678 = 68AC$$

$$68AC + 9ABC = 10368$$

End-around carry:

$$0368 + 1 = 0369$$

Complement:

$$\boxed{FC96}$$

37.1.2 Problem 2

Given 1111, 2222, 3333, 4444:

$$1111 + 2222 + 3333 + 4444 = AAAA$$

$$\boxed{\text{Checksum} = 5555}$$

Verification:

$$AAAA + 5555 = FFFF.$$

37.1.3 Problem 3

Given ABCD, 1234, FFFF:

$$ABCD + 1234 + FFFF = 1BE00$$

Fold carry:

$$BE00 + 1 = BE01$$

Complement:

$$\boxed{41FE}$$

Verification:

$$BE01 + 41FE = FFFF.$$

37.2 B. Even and Odd Parity — Cross-Unit Revision

Parity Revision — Even and Odd Parity



Parity is primarily a Data Link Layer error-detection topic; included here for exam-oriented cross-unit revision.

Note: Parity belongs primarily to the Data Link Layer / Unit II, but it is included here as an examination-oriented revision topic because error detection is closely related to transport-layer error control. Do not confuse parity with checksum or CRC.

37.2.1 Problem 4 — Even Parity

Data:

1011001

Number of 1s = 4, already even. Therefore:

$$\boxed{\text{Even parity bit} = 0}$$

Transmitted sequence (parity appended):

10110010.

37.2.2 Problem 5 — Odd Parity

The same data has four 1s. To make the total odd:

Odd parity bit = 1

Transmitted sequence:

10110011.

37.2.3 Problem 6 — Detect an Error

Suppose a receiver expects even parity but receives 10110011. The received sequence contains five 1s, which is odd. Therefore the parity check fails and an error is detected.

Common Mistake

A parity check can detect many single-bit errors but cannot guarantee detection of all multiple-bit error patterns. It is much simpler than checksum or CRC and should not be described as a replacement for them.

37.3 C. TCP Sequence and ACK Problems

37.3.1 Problem 7

A TCP segment has Seq = 1000 and carries 500 bytes. Find the next ACK if all bytes are received correctly.

$$1000 + 500 = 1500$$

ACK = 1500

37.3.2 Problem 8

A segment has Seq = 5000 and length 1000 bytes. The next segment begins at 6000 and carries 700 bytes. Find the ACK after both arrive in order.

$$5000 + 1000 = 6000$$

$$6000 + 700 = 6700$$

ACK = 6700

37.3.3 Problem 9 — SYN

A client sends SYN with Seq = 3000. What ACK does the server return for the SYN? A SYN consumes one sequence number:

ACK = 3001

37.4 D. TCP Window / Flow-Control Problems

37.4.1 Problem 10

Given $rwnd = 5000$ bytes and $cwnd = 8000$ bytes:

$$\min(5000, 8000) = 5000$$

Usable limit = 5000 bytes

37.4.2 Problem 11

If $rwnd = 9000$ and $cwnd = 6000$:

Usable limit = 6000 bytes

The network-side congestion window is the limiting factor.

37.5 E. Simple Congestion-Window Problem

37.5.1 Problem 12

Use the simplified classroom model with $cwnd = 1$ MSS and $ssthresh = 8$ MSS. During slow start:

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 8.$$

Then congestion avoidance grows approximately one MSS per RTT:

$$9 \rightarrow 10 \rightarrow 11 \rightarrow \dots$$

This is an illustrative classical model, not a universal description of every modern TCP implementation.

37.6 F. Important Comparisons

Comparison	First concept	Second concept
UDP vs TCP	Connectionless, no built-in reliable byte stream	Connection-oriented reliable byte stream
Flow vs Congestion control	Protects receiver	Protects network
Error vs Congestion control	Detects/repairs delivery errors/loss	Regulates injection rate under network congestion
Symmetric vs Asymmetric Encryption vs Hashing	Shared secret key Confidentiality	Public/private key pair Digest/integrity-related applications
Digital signature vs Encryption	Integrity/authentication	Confidentiality
Authentication vs Authorization	Establish identity/control	Decide permitted actions
HTTP vs HTTPS	HTTP application protocol	HTTP protected with TLS
CIA	Confidentiality, Integrity, Availability	Three complementary security objectives

37.7 G. Unit IV Last-Minute Revision

Transport Layer

- Process-to-process delivery uses transport-layer port numbers.

- UDP header: Source Port, Destination Port, Length, Checksum.
- UDP header size = 8 bytes.
- UDP checksum uses one's-complement arithmetic and a pseudo-header.
- TCP = connection-oriented, reliable, ordered byte-stream service.
- TCP header includes ports, sequence number, acknowledgment number, data offset, flags, window, checksum, urgent pointer, options and data.
- Three-way handshake: SYN → SYN-ACK → ACK.
- Orderly termination: FIN/ACK in each direction.
- Flow control protects the receiver; congestion control protects the network.
- TCP error control uses checksum, sequence numbers, ACKs and retransmission.
- Classical congestion control: slow start, congestion avoidance, fast retransmit, fast recovery.

QoS

Bandwidth, delay, jitter, packet loss and reliability are core parameters. Techniques include traffic shaping, scheduling, admission control, resource reservation, classification/marketing and capacity planning.

Security

CIA = Confidentiality, Integrity, Availability. Symmetric cryptography uses shared secret key material. Asymmetric cryptography uses public/private keys. Hashes produce digests. MACs use secret keys for message authentication. Digital signatures use private-key signing and public-key verification. Authentication is not authorization. HTTPS uses TLS. E-mail security can use S/MIME/OpenPGP mechanisms. Security also has privacy, ethical and social dimensions.

37.8 One-Page Quick-Revision Cheat Sheet

Important Exam Point

UDP: 8-byte header; ports + length + checksum; connectionless; no TCP-style reliability/order/flow control/congestion control.

TCP: reliable ordered byte stream; 16-bit ports; 32-bit sequence and ACK numbers; receiver window; checksum; current header flags include CWR, ECE, URG, ACK, PSH, RST, SYN, FIN.

Handshake: SYN(x), SYN-ACK(y, ACK x+1), ACK(y+1).

ACK rule: ACK number = next sequence number expected.

Flow control: receiver capacity → *rwnd*.

Congestion control: network capacity → *cwnd*; usable limit approximately $\min(rwnd, cwnd)$.

Checksum: add 16-bit words using one's-complement addition; fold end-around carry; complement. Receiver check → *FFFF*.

QoS: bandwidth, delay, jitter, loss, reliability.

CIA: confidentiality, integrity, availability.

Crypto: symmetric = shared secret; asymmetric = public/private.

Signature: sign with private key; verify with public key; supports integrity and signer authentication.

Authentication: establish claimant control of authenticator.

HTTPS: HTTP over TLS.

Parity: cross-unit revision only; do not confuse with checksum/CRC.

37.9 Final Exam Strategy

For a 10-mark transport/security answer, begin with a precise definition, draw the topic-specific diagram, explain fields/steps in logical order, include one concrete example or numerical calculation where applicable, state advantages/limitations or security properties, and conclude with the main purpose.

Part V

UNIT V – APPLICATION LAYER

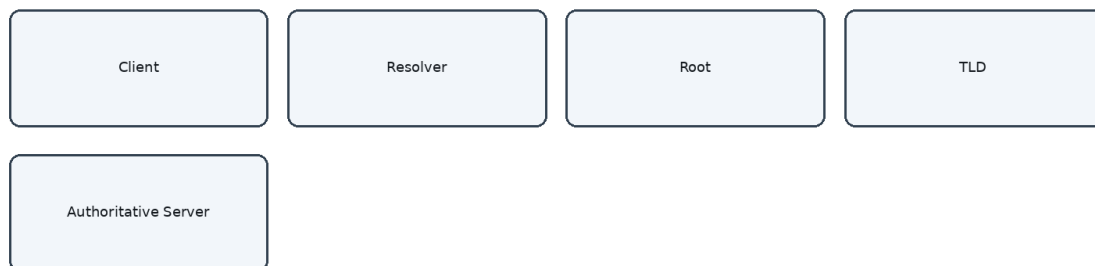
Chapter 38

Domain Name System (DNS): Distribution of Name Space and DNS in the Internet

Definition

Learning objectives: Explain DNS hierarchy, name resolution, records, caching and Internet operation.

DNS Resolution Concept



38.1 Purpose

DNS maps human-readable domain names to resource records used by network applications. It is a distributed, hierarchical naming system.

38.2 Name Space

The hierarchy has a root, top-level domains and delegated authoritative zones. A domain may contain subdomains and host/service records.

38.3 Resolution

A client normally queries a recursive resolver. If the answer is not cached, the resolver may query authoritative infrastructure through the hierarchy and return the result to the client.

38.4 Common Records

A maps a name to an IPv4 address; AAAA maps a name to an IPv6 address; MX identifies mail exchange servers; CNAME provides an alias; NS identifies authoritative name servers.

38.5 Caching

Resolvers cache records according to TTLs, reducing repeated queries and improving performance.

2-Mark Answer

DNS is a distributed hierarchical system for resolving domain names and other resource records.

5-Mark Answer

Explain DNS hierarchy and resolution.

10-Mark Answer

Explain DNS name space, Internet DNS architecture, resolution process, common records and caching.

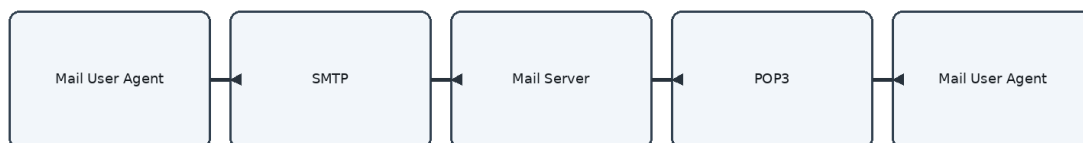
Chapter 39

Electronic Mail: Architecture and Services

Definition

Learning objectives: Explain e-mail architecture, user agents, mail servers and basic message delivery.

Electronic Mail Architecture



39.1 Architecture

Electronic mail involves user agents, mail submission/transfer components, mailboxes and retrieval/access mechanisms. SMTP is used for message transfer/submission; POP3 is one of the protocols used to retrieve mail.

39.2 Services

Typical services include composing, addressing, submission, transfer, storage and retrieval.

39.3 Message Path

A sender's mail user agent submits a message to a mail service. Mail transfer systems use SMTP to relay it toward the recipient's mail system, where it is stored for later access by the recipient.

Important Exam Point

SMTP is a mail transfer/submission protocol; POP3 is primarily a mailbox retrieval protocol. Do not interchange their roles.

2-Mark Answer

SMTP is used for e-mail transfer/submission; POP3 is used for mailbox retrieval.

5-Mark Answer

Explain e-mail architecture and the roles of user agents, mail servers and SMTP/POP3.

10-Mark Answer

Describe e-mail architecture and services with a labeled message-flow diagram.

Chapter 40

POP3 and SMTP

Definition

Learning objectives: Explain the basic functions and distinction between POP3 and SMTP.

40.1 SMTP

Simple Mail Transfer Protocol is an application-layer protocol used to transfer mail between mail systems and for message submission in appropriate deployments.

40.2 POP3

Post Office Protocol version 3 provides a simple mechanism for a client to retrieve messages from a maildrop. It commonly uses a download-oriented model; modern deployments may provide server-side synchronization through other protocols such as IMAP, but IMAP is outside this syllabus.

	SMTP	POP3
Primary role	Send/transfer mail	Retrieve mail
Direction	Sender/client to mail server or server to server	Mail server to recipient client
Typical concept	Store-and-forward transfer	Maildrop retrieval

2-Mark Answer

SMTP sends/transfers e-mail; POP3 retrieves e-mail from a maildrop.

5-Mark Answer

Differentiate SMTP and POP3.

10-Mark Answer

Explain SMTP and POP3 operation and their roles in e-mail architecture.

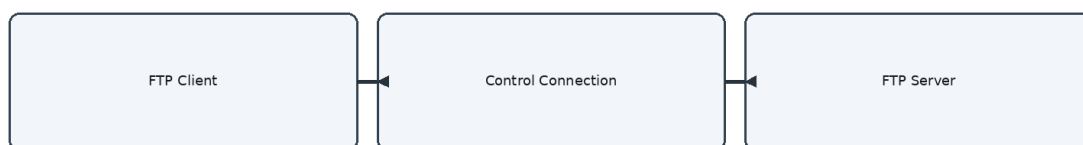
Chapter 41

File Transfer Protocol (FTP)

Definition

Learning objectives: Explain FTP architecture, control/data concepts and basic operations.

FTP Concept



41.1 Purpose

FTP is an application-layer protocol for transferring files between a client and server. Classic FTP separates control communication from data transfer.

41.2 Basic Operations

Users authenticate as required, navigate directories, upload/download files and terminate the session. Commands are carried over the control connection; file/listing transfers use the data connection.

41.3 Security Note

Classic FTP does not by itself provide modern cryptographic confidentiality. Secure file-transfer deployments commonly use other mechanisms, but the syllabus topic here is FTP itself.

2-Mark Answer

FTP is an application-layer protocol for transferring files between hosts.

5-Mark Answer

Explain FTP architecture and the control/data separation.

10-Mark Answer

Explain FTP operation, connections, common operations and security limitations.

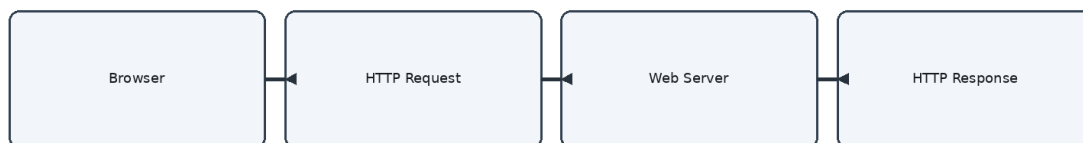
Chapter 42

World Wide Web and HTTP

Definition

Learning objectives: Explain the WWW, client-server model, HTTP request/response and basic methods/status concepts.

HTTP / WWW Concept



42.1 WWW

The World Wide Web is a distributed hypermedia system in which clients retrieve resources from web servers using application protocols such as HTTP.

42.2 HTTP

HTTP is a request/response application protocol. A client sends a request identifying a resource and desired operation; the server returns a response containing a status code and, when applicable, a representation of the resource.

42.3 Common Methods

GET retrieves a representation; POST submits data for processing; PUT replaces/creates a representation at a target resource when supported; DELETE requests deletion according to application semantics.

42.4 Status Codes

1xx informational, 2xx successful, 3xx redirection, 4xx client-error, 5xx server-error classes.

42.5 HTTP vs HTTPS

HTTPS is HTTP carried over TLS, providing cryptographic protection of the connection.

2-Mark Answer

HTTP is a request/response protocol used for web resource transfer.

5-Mark Answer

Explain HTTP request and response with common methods/status classes.

10-Mark Answer

Explain WWW architecture, HTTP operation, methods, status codes and HTTPS concept.

Part VI

EXAMINATION SUPPORT

Complete MCA 301 Examination Question Bank

2-Mark Questions

1. Define data communication.
2. What is simplex transmission?
3. Define attenuation.
4. What is multiplexing?
5. What is the purpose of the OSI model?
6. Define framing.
7. What is parity?
8. What is CRC?
9. What is ARQ?
10. Define Go-Back-N.
11. What is an IPv4 address?
12. What is IPv6?
13. What is ARP?
14. What is DHCP?
15. What is ICMP?
16. Define routing.
17. What is process-to-process delivery?
18. List the fields of a UDP header.
19. What is a TCP sequence number?
20. What is flow control?
21. What is congestion control?
22. Define QoS.
23. State the CIA triad.
24. What is symmetric cryptography?
25. What is a digital signature?
26. Define authentication.

27. What is DNS?
28. What is SMTP?
29. What is FTP?
30. What is HTTP?

5-Mark Questions

1. Explain the components of data communication.
2. Compare LAN, MAN and WAN.
3. Explain bus, star and mesh topologies.
4. Explain the OSI layers.
5. Explain transmission impairments.
6. Explain FDM, WDM and TDM.
7. Compare circuit and packet switching.
8. Explain framing and flow control.
9. Explain parity, checksum and CRC.
10. Explain Stop-and-Wait ARQ.
11. Compare Go-Back-N and Selective Repeat.
12. Explain IPv4 and IPv6 notation.
13. Explain ARP and DHCP.
14. Compare distance-vector and link-state routing.
15. Explain ICMP and IGMP.
16. Explain UDP header fields.
17. Explain TCP services.
18. Explain TCP three-way handshake.
19. Explain TCP flow control.
20. Explain QoS parameters.
21. Explain the CIA triad.
22. Compare symmetric and asymmetric cryptography.
23. Explain digital signatures.
24. Explain authentication factors.
25. Explain DNS resolution.

10-Mark Questions

1. Explain data communication, components, representation and data flow.
2. Explain OSI and TCP/IP models with comparison.
3. Explain network topologies in detail.

4. Explain physical-layer signals, impairments and performance measures.
5. Explain multiplexing and transmission media.
6. Explain circuit switching and packet switching.
7. Explain framing, flow control and error control at the data-link layer.
8. Explain parity, checksum, CRC and Hamming code with numerical examples.
9. Explain Stop-and-Wait, Go-Back-N and Selective Repeat ARQ.
10. Explain IPv4/IPv6 addressing and compare the protocols.
11. Explain IPv4-to-IPv6 transition mechanisms.
12. Explain ARP, RARP and DHCP.
13. Explain distance-vector and link-state routing with RIP and OSPF.
14. Explain multicast routing, DVMRP and MOSPF.
15. Explain process-to-process delivery, ports, multiplexing and demultiplexing.
16. Explain UDP format, operations and checksum.
17. Explain TCP segment format and connection management.
18. Explain TCP flow, error and congestion control.
19. Explain QoS parameters and techniques to improve QoS.
20. Explain network security and the CIA triad.
21. Compare symmetric and asymmetric cryptography.
22. Explain message security and digital signatures.
23. Explain user authentication, e-mail security and web security.
24. Explain DNS and the Internet e-mail architecture.
25. Explain FTP, WWW and HTTP.

Numerical Problems

1. Calculate transmission and propagation delay for a given packet, rate, distance and propagation speed.
2. Solve an even-parity problem.
3. Solve an odd-parity problem.
4. Generate and verify a one's-complement checksum.
5. Solve a CRC using a specified generator.
6. Determine Hamming-code parity positions for a given number of data bits.
7. Solve a Stop-and-Wait efficiency problem using a stated model.
8. Determine the TCP acknowledgment number for a segment with a given sequence number and payload length.
9. Determine the usable TCP sending window from rwnd and cwnd.
10. Solve a simple classical TCP congestion-window growth problem.

MCA 301 – Complete Last-Minute Revision

Unit I

- Data communication: sender, receiver, message, medium, protocol.
- Data flow: simplex, half-duplex, full-duplex.
- Criteria: performance, reliability, security.
- OSI: Physical, Data Link, Network, Transport, Session, Presentation, Application.
- Impairments: attenuation, distortion, noise.
- Delays: transmission = packet bits/rate; propagation = distance/speed.
- Multiplexing: FDM, WDM, TDM.
- Media: twisted pair, coaxial, fiber; radio, microwave, infrared.
- Switching: circuit and packet.

Unit II

- Framing; flow control.
- Error detection: parity, checksum, CRC; correction: Hamming code.
- ARQ: Stop-and-Wait, Go-Back-N, Selective Repeat.
- Wireless LAN: IEEE 802.11; Bluetooth.
- Virtual-circuit networks: Frame Relay, ATM; ATM cell = 53 bytes.

Unit III

- IPv4 = 32 bits; IPv6 = 128 bits.
- IPv6 base header = 40 bytes.
- Dual stack = both IPv4 and IPv6; tunneling = one protocol carried inside another.
- ARP resolves IPv4 to link-layer address; DHCP provides configuration; RARP is historical.
- ICMP = IP control/diagnostics; IGMP = IPv4 multicast membership.
- Distance vector: RIP; link state: OSPF.
- DVMRP = multicast distance-vector family; MOSPF = multicast extension of OSPF concepts.

Unit IV

- UDP header = source port, destination port, length, checksum.
- TCP = reliable ordered byte stream.
- TCP setup = SYN, SYN-ACK, ACK.
- ACK = next sequence number expected.
- Flow control = receiver protection (rwnd).
- Congestion control = network protection (cwnd).
- QoS = bandwidth, delay, jitter, loss, reliability.
- CIA = confidentiality, integrity, availability.
- Symmetric = shared secret; asymmetric = public/private.
- Signature = private-key signing, public-key verification.
- Authentication $\neq$ authorization.

Unit V

- DNS = distributed hierarchical naming.
- SMTP = mail transfer/submission; POP3 = mail retrieval.
- FTP = file transfer with separate control/data concepts.
- HTTP = request/response web protocol; HTTPS = HTTP over TLS.

High-Value Comparison Rules

Comparison	First	Second
Propagation vs Transmission delay	Distance/speed	Bits/rate
Circuit vs Packet switching	Dedicated path/resources	Shared packet network
GBN vs SR	Retransmit from loss onward	Retransmit selected missing frames
IPv4 vs IPv6	32-bit	128-bit
ARP vs DHCP	Address-to-link resolution	Host configuration
UDP vs TCP	Datagram/low mechanism	Reliable ordered byte stream
Flow vs Congestion control	Receiver	Network
Symmetric vs Asymmetric	Shared secret	Public/private
Authentication vs Authorization	Establish identity/control	Determine permissions
HTTP vs HTTPS	HTTP	HTTP protected by TLS

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